



ArdaLink

PASTORALIST INTELLIGENCE PLATFORM

JHUB AFRICA · 2026

Pastoralist intelligence for Isiolo's drylands.

SATELLITE · AI · HERDER'S PHONE · VOICE · USSD · SMS

THE PROBLEM

A herder loses KES 200,000 of livestock in a bad drought cycle.

Drought is the single largest cause of household poverty in northern Kenya. Satellites see the vegetation stress two to three weeks before a herder does — but the signal never reaches the phone.

BY THE NUMBERS

KES 200K lost per herder in a bad cycle

2–3 wks satellite lead vs. herder's eye

THE SOLUTION

A voice-first intelligence layer that lives on a 2G phone.

Every five days, the satellite reads every patch of the ward. When the pasture starts to fail, the platform calls the herder in Swahili, English, or Borana to ask what she sees on the ground.

VOICE

Outbound + inbound call

USSD

*123*8# menu, no data plan

SMS

BULA · MALISHO · ONGEA

ARCHITECTURE

Three small services. One secure database. That's the whole platform.

HERDER

2G/3G phone

- Voice call
- (Swahili / English)
- USSD *123*8#
- SMS keyword
- (BULA · MALISHO)

AFRICA'S TALKING

Voice · SMS · USSD

- Outbound voice API
- 2-way SMS gateway
- USSD gateway
- DID pool
- +254 20 5XX XXXX

ARDALINK CORE

3 services

- The watcher
- satellite + weather
- The brain
- composes the brief
- The voice
- calls her in Swahili
- The dashboard
- what operators see

DATA + AI

satellite · language · store

- Satellite reads the ward
- every 5 days
- AI spots trouble early
- in her language
- Her answers are saved
- to the database

→ EVERY 5 DAYS · satellite reads the ward → AI writes the brief → we call her → she decides

THE HERDER JOURNEY

From dial-tone to ground-truth report in five steps.

1 DIAL

Herder dials *123*8#

2 PICK

1 = Bula Pesa
2 = Malisho
3 = Ongea

3 REPLY

ArdaLink replies in Swahili or English

4 CALL

Outbound voice call: AI brief in her dialect

5 REPORT

Herder answers. Indicators saved to ground truth

WHAT'S BUILT · WHAT'S NEXT

3

demo wards

bula-pesa · garbatulla · merti

134

tests pass

api + web + engine

\$0.12

per herder / month

at 50K-household scale

6 wks

to first call

after green light

4

AT webhooks

voice · USSD · SMS · events

NEXT · ● Africa's Talking production key + DID ● AI voice stack (real-time speech) ● Earth Engine satellite account ● Recording voices in 4 local dialects



github.com/JHUB-AFRICA/arda-link-ai

Source of truth · architecture · sandbox workflow. This poster is generated from the same repo.

THE TEAM

Founder · Ops · Domain · Software developer

JHUB Africa · Isiolo County